COLLEGE OF
ENGINEERING AND
COMPUTER SCIENCE

CCDS-322 Project

Traffic Congestion Prediction
using Machine Learning

Presented To
Ines Boufateh
Presented by
Rimas Albishri 2310996
Remas Alzhrani 2310987
Fatima Salem 2311048

Traffic Congestion Prediction

Introduction

Urban Traffic Congestion

Rapid population and vehicle growth increase traffic congestion in cities.

Efficient Traffic Prediction

Accurate prediction helps improve transportation systems and reduce delays.

Machine Learning Application

Machine learning can analyze traffic data and predict congestion levels.

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Project Goal

Goal

Develop machine learning models to classify traffic congestion levels into Low, Medium, and High

Purpose

- Support decision-making in traffic management.
- Reduce traffic delays and emissions.
- Improve urban mobility.

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Dataset Description and Rationale

The dataset used in this study is titled “IoT Traffic Dataset” and was obtained from Kaggle.

Column Name	Description
Latitude, Longitude	Geographic coordinates of the location
Traffic_Speed_kmph	Traffic speed in kilometers per hour
Traffic_Density_vpkm	Traffic density
FFT_Feature_1,FFT_Feature_2, FFT_Feature_3	Features extracted using Fast Fourier Transform (FFT).
Vehicle_Count	Number of vehicles in the area
Time_of_Day_min	Time in minutes of the day
Congestion_Level	Level of congestion (0: Low, 1: Medium, 2: High)

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Research Hypotheses and Questions

Hypothesis 1: Higher traffic density leads to higher congestion levels.
Hypothesis 2: Lower traffic speeds indicate higher congestion.

Research Questions:

- How do traffic speed and density affect the congestion level?
- Can predictive models provide accurate predictions for different congestion levels?
- Which machine learning algorithm performs best for this prediction task?

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Tools and Libraries

Python (Google Colab)

Libraries Used:

- Pandas
- NumPy
- Matplotlib & Seaborn
- Scikit-learn

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Data Preparation

Before applying machine learning models, the dataset was cleaned and prepared to improve model performance.

Steps:

- Checked missing values: none were found.
- Checked class distribution.
- Selected relevant features for training.
- Split the data into 80% training and 20% testing.
- Used stratified sampling to keep class proportions.
- Applied StandardScaler for feature scaling.

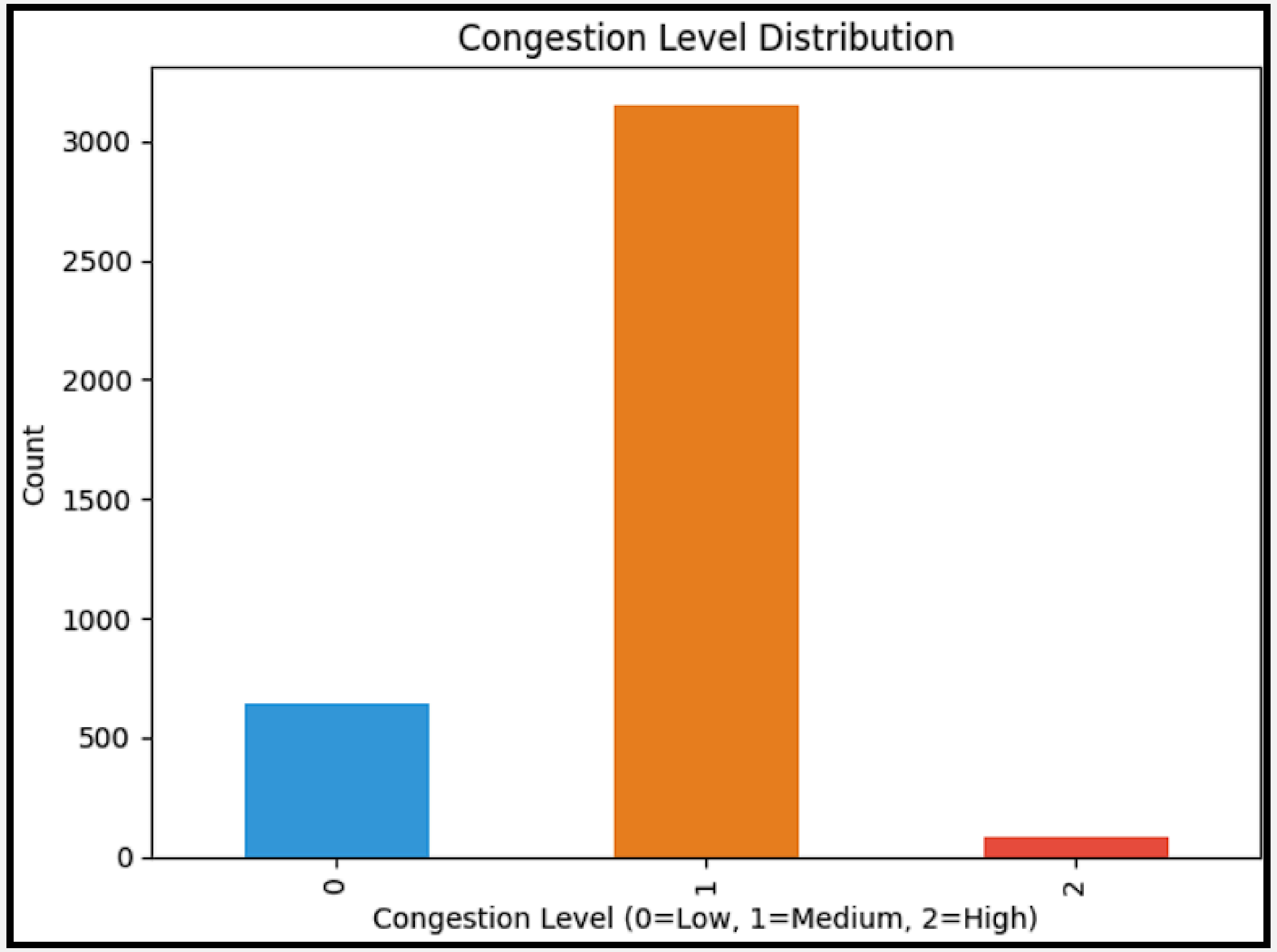

Figure: Congestion Level Distribution

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Machine Learning Models and Results

Three supervised machine learning models were applied:

- Logistic Regression: baseline linear model
- Decision Tree: captures non-linear decision rules
- Random Forest: ensemble model using multiple trees
- Best Model: Random Forest
- Test Accuracy: 81.19%
- CV Accuracy: 81.16%

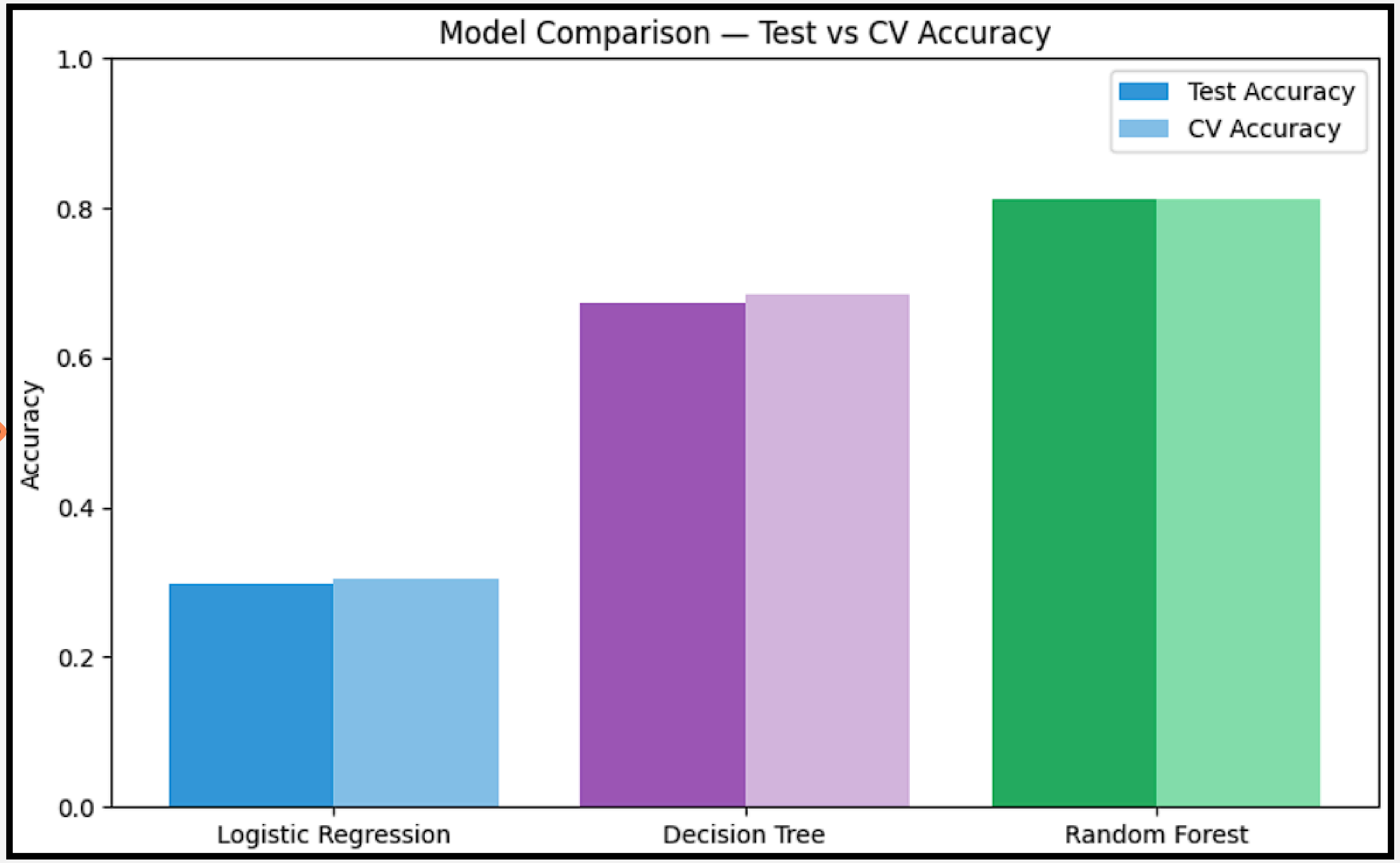

Figure: Test Accuracy vs Cross-Validation Accuracy

Best Model and Limitation

Best Model: Random Forest

Performance:

- **Test Accuracy:** 81.19%
- **CV Accuracy:** 81.16%

Main Limitation:

Although Random Forest achieved the highest overall accuracy, it struggled to detect the High congestion class due to the small number of High congestion samples.

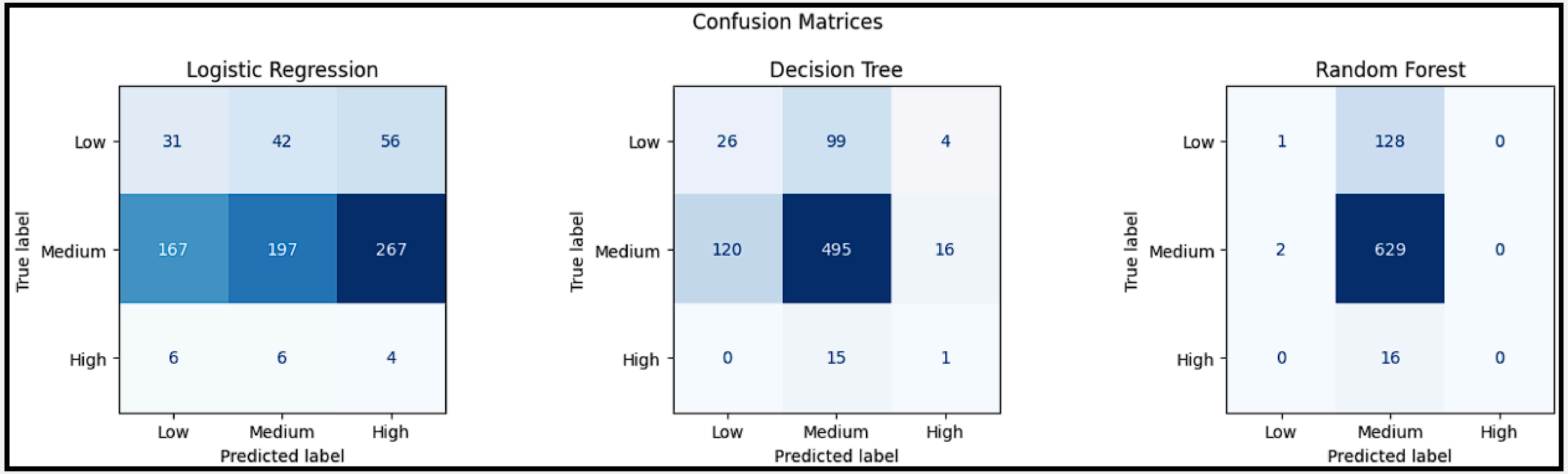

Figure: Confusion Matrices for All Models

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Ethical and Professional Responsibilities

Machine learning systems for traffic prediction must be developed responsibly.

- **Data Privacy:** Traffic and location data should be protected.
- **Bias and Fairness:** Imbalanced data can cause biased predictions.
- **Transparency:** Model results should be understandable.
- **Reliability and Safety:** Predictions should be monitored carefully.
- **Accountability:** Human supervision is still needed.

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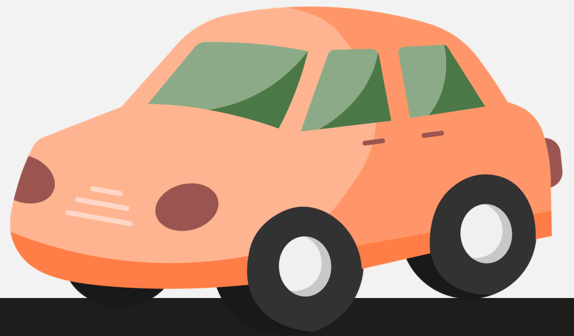

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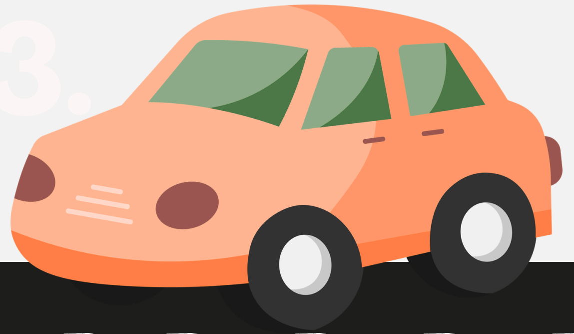
Recommendations and Future Work

To improve the project in the future:

- Use SMOTE to handle class imbalance.
- Collect more High congestion samples.
- Add real-world features such as weather, accidents, and public events.
- Apply Grid Search for hyperparameter tuning.
- Test advanced models such as XGBoost or Neural Networks.
- Build a real-time traffic dashboard.



Conclusion and References

This project showed that machine learning can be used to predict traffic congestion levels using IoT traffic data.

Main Findings:

- Three models were tested.
- Random Forest achieved the best performance.
- The highest test accuracy was 81.19%.
- Class imbalance was the main limitation.
- Future work should improve High congestion detection.

References

- Kaggle. IoT Traffic Dataset.
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